Ethanol Blending in Gasoline – Costa Rica

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U.S. GRAINS & BIOPRODUCTS COUNCIL

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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.

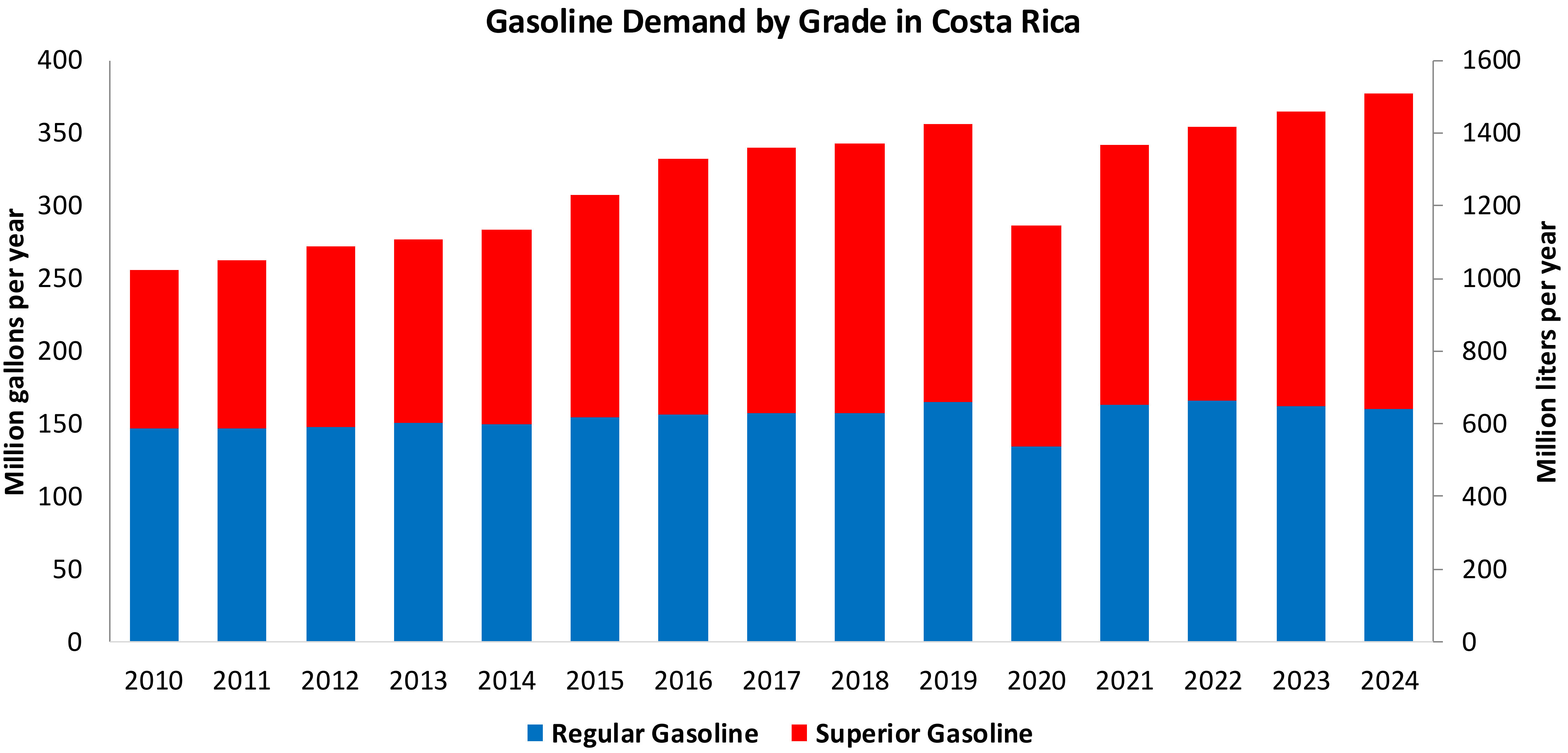
Ethanol Blending in Gasoline - Costa Rica

In 2024, gasoline demand was 378 million gallons (1429 million liters). Costa Rica has two gasoline RON 95 and RON 91, with a market share of 45% for RON 91 and 55% for RON 95. The Limon refinery stopped production in 2012. Gasoline supply is done with imports mainly from the United States and some from Europe.

Gasoline blends with up to 10% v/v of ethanol are allowed (INTE E1:2019). However, this mandate is not yet reflected on the gasoline market. Current ethanol consumption is less than 0.2% because there is no stakeholder’s consensus for implementation. In 2024, ethanol production was 8.9 million gallons (33 million liters) and ethanol exports mainly to Europe was 7.2 million gallons (27 million liters). Introduction of E10 is expected by the Government in 2026.

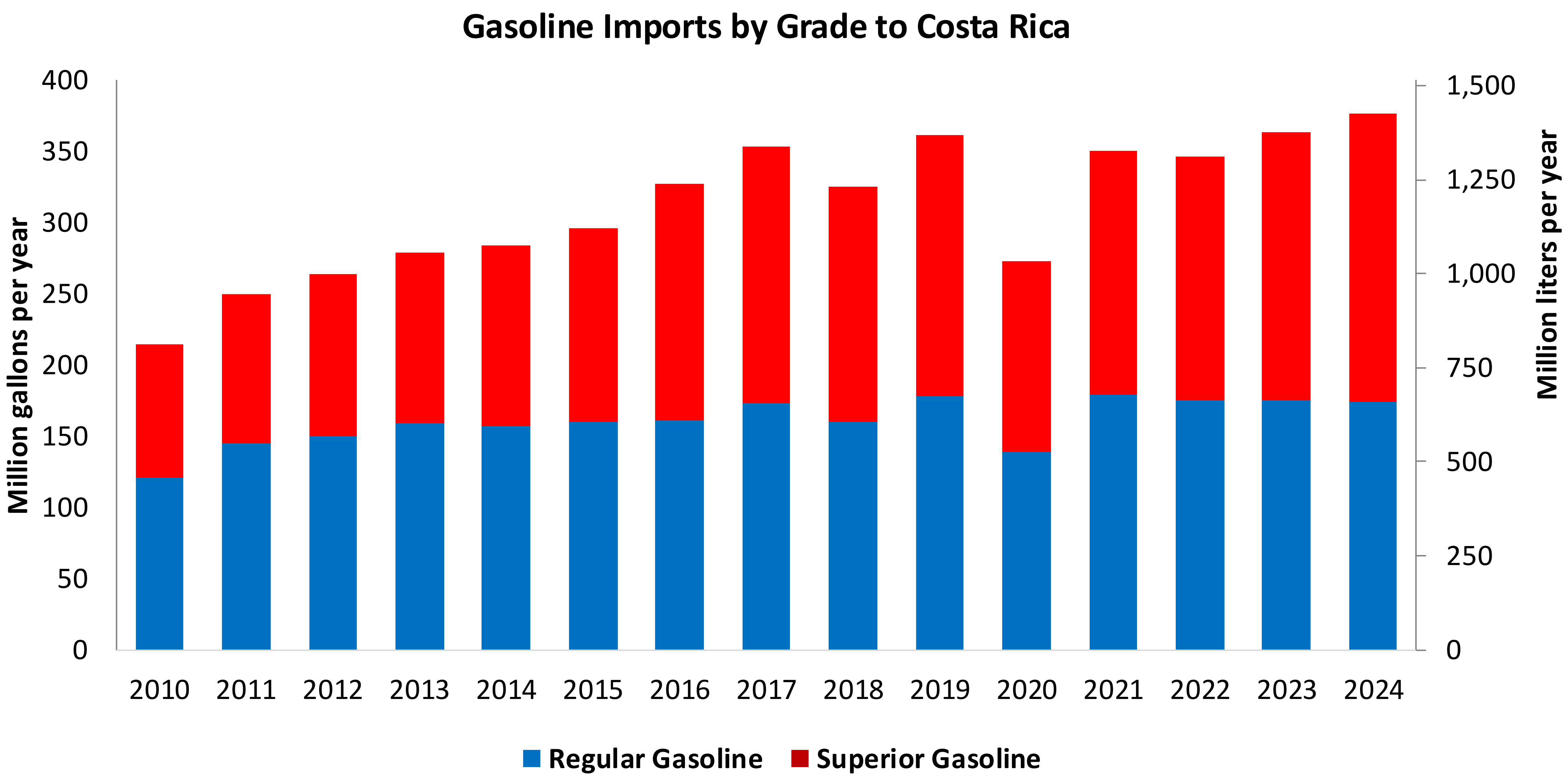
Source: Sepse

Gasoline Demand in Costa Rica



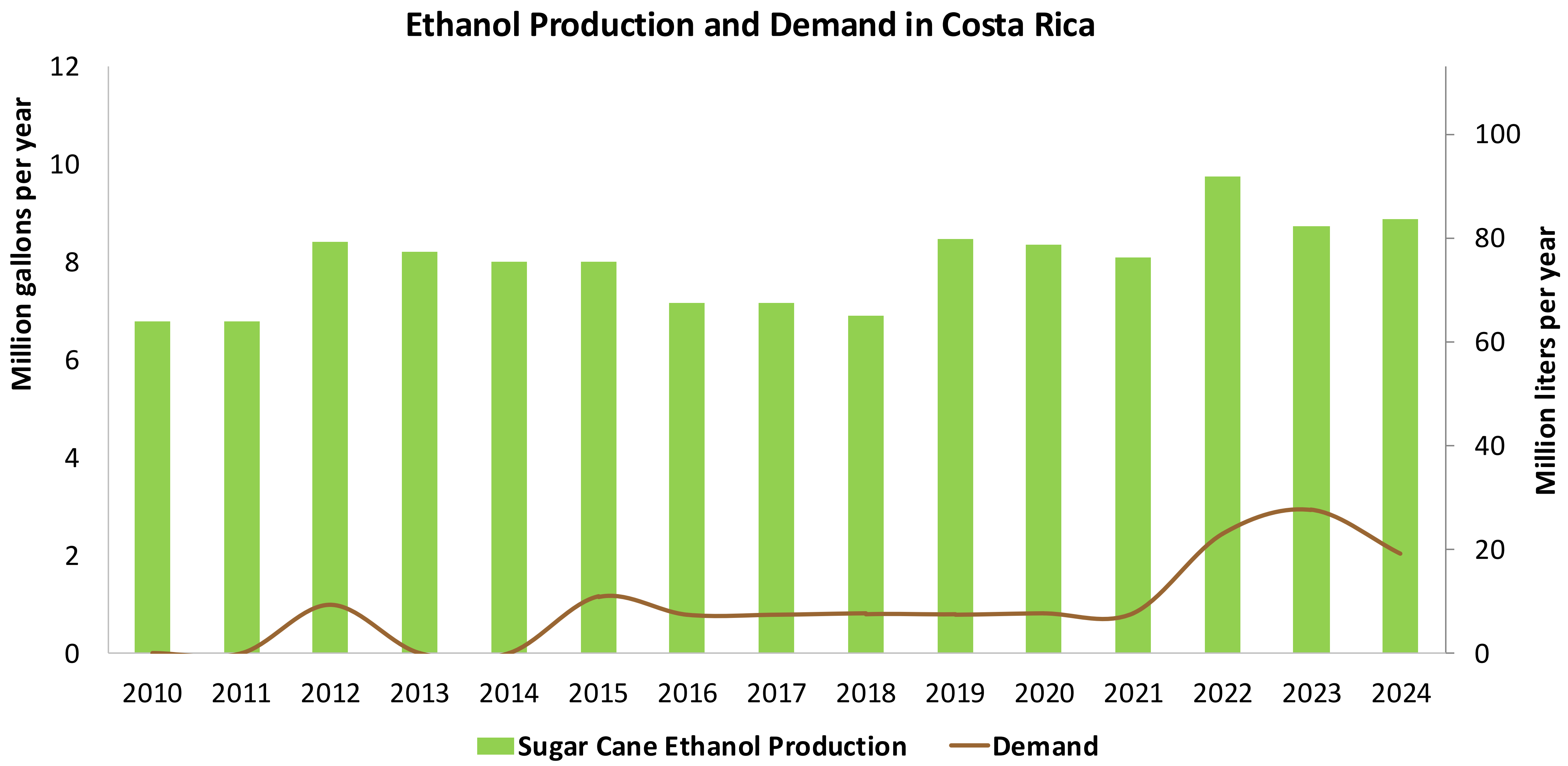
Source: Sepse

Gasoline Imports in Costa Rica



Source: Sepse

Ethanol Demand in Costa Rica



Source: Sepse

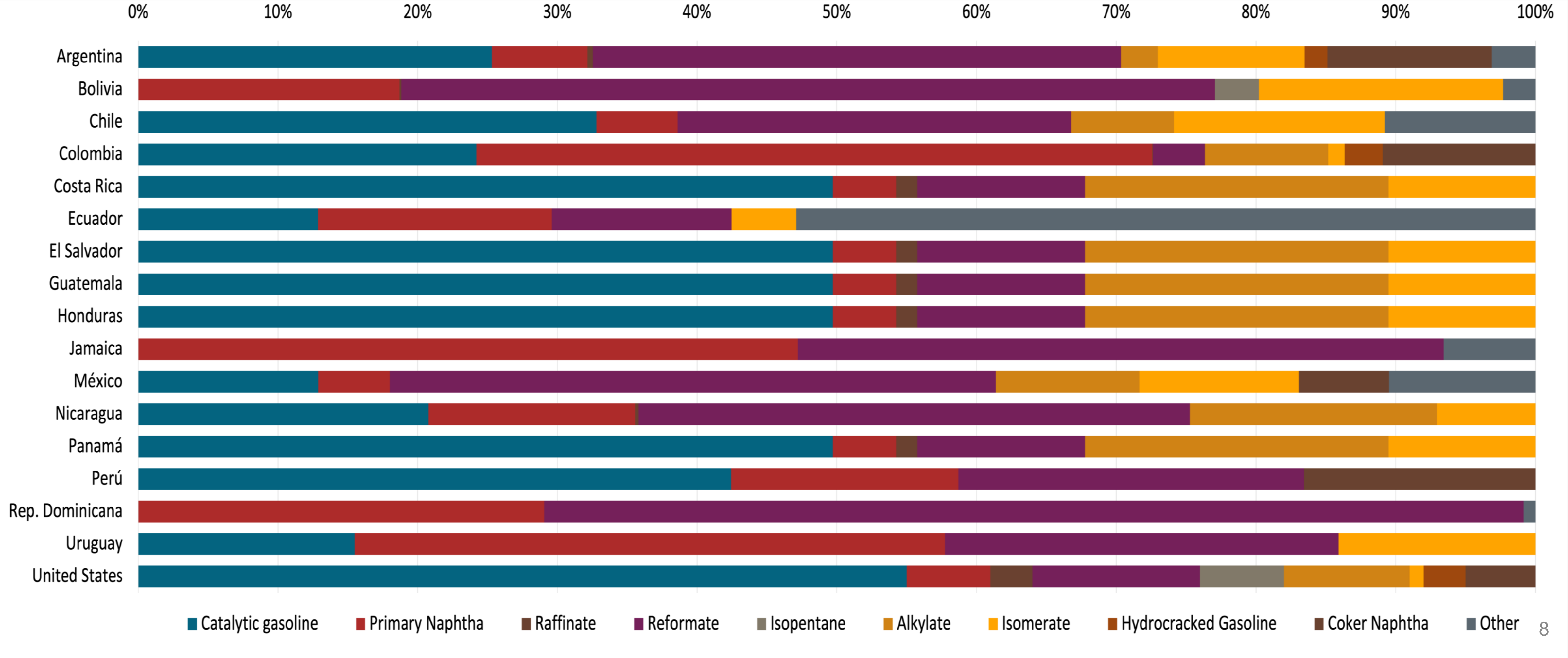
Gasoline Quality in Costa Rica

Name	INTE E1:2019		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2019		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	RON 91	RON 95	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1,5% v/v	< 1,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 35% v/v	< 35% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 18% v/v	< 18% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	> 79	> 83	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 50 mg/kg	< 50 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	2,7% m/m (3,7% m/m if ethanol is added)	2,7% m/m (3,7% m/m if ethanol is added)	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 10% v/v	< 10% v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa (< 76 kPa if ethanol is added)	< 69 kPa (< 76 kPa if ethanol is added)	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-		-	-	-	-
Ethers 5 or more C Atoms	-		Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: INTECO

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



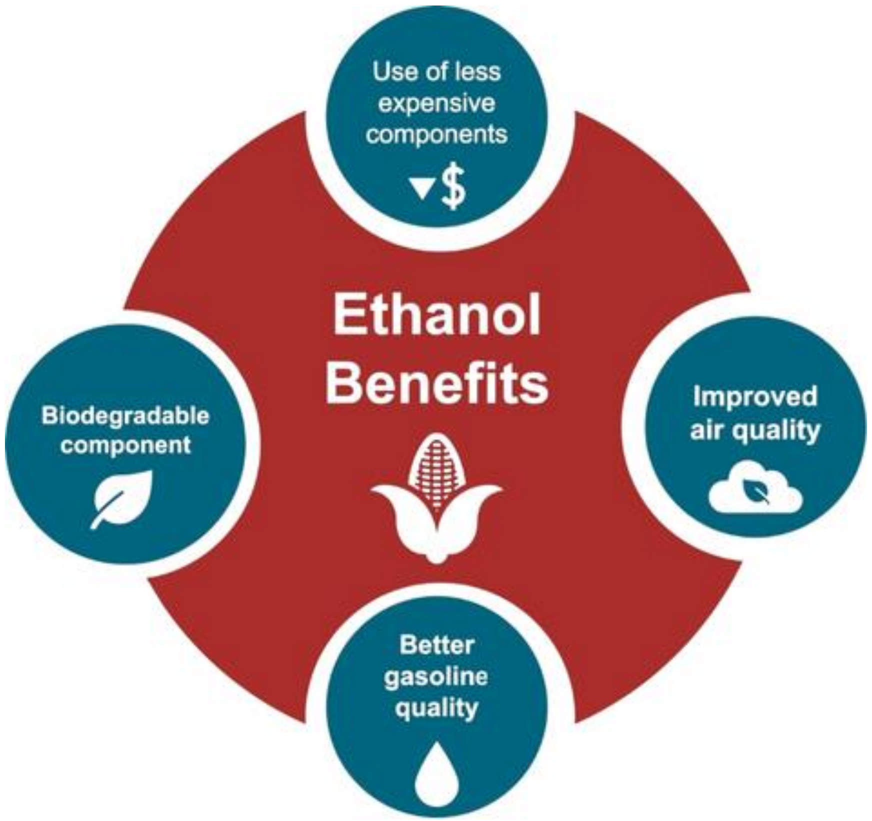
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

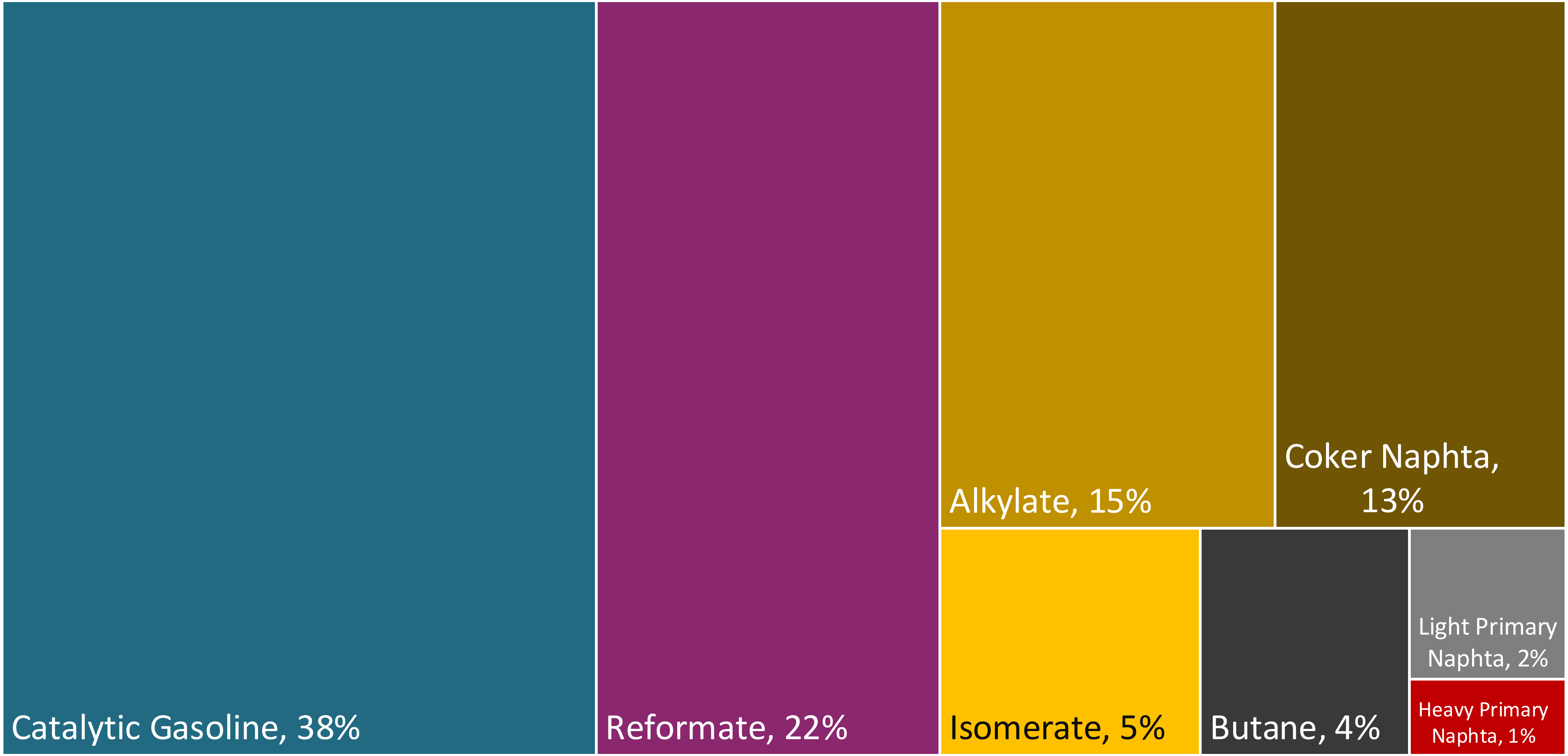
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

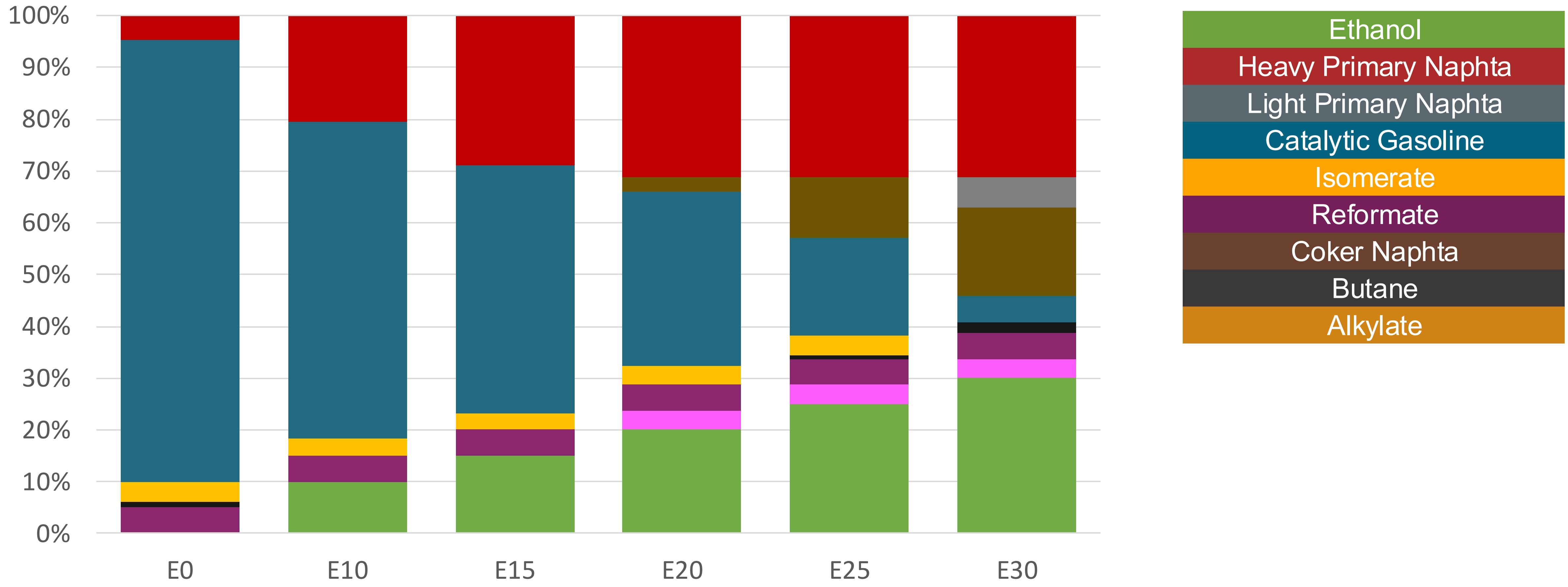


Blending Components - Costa Rica



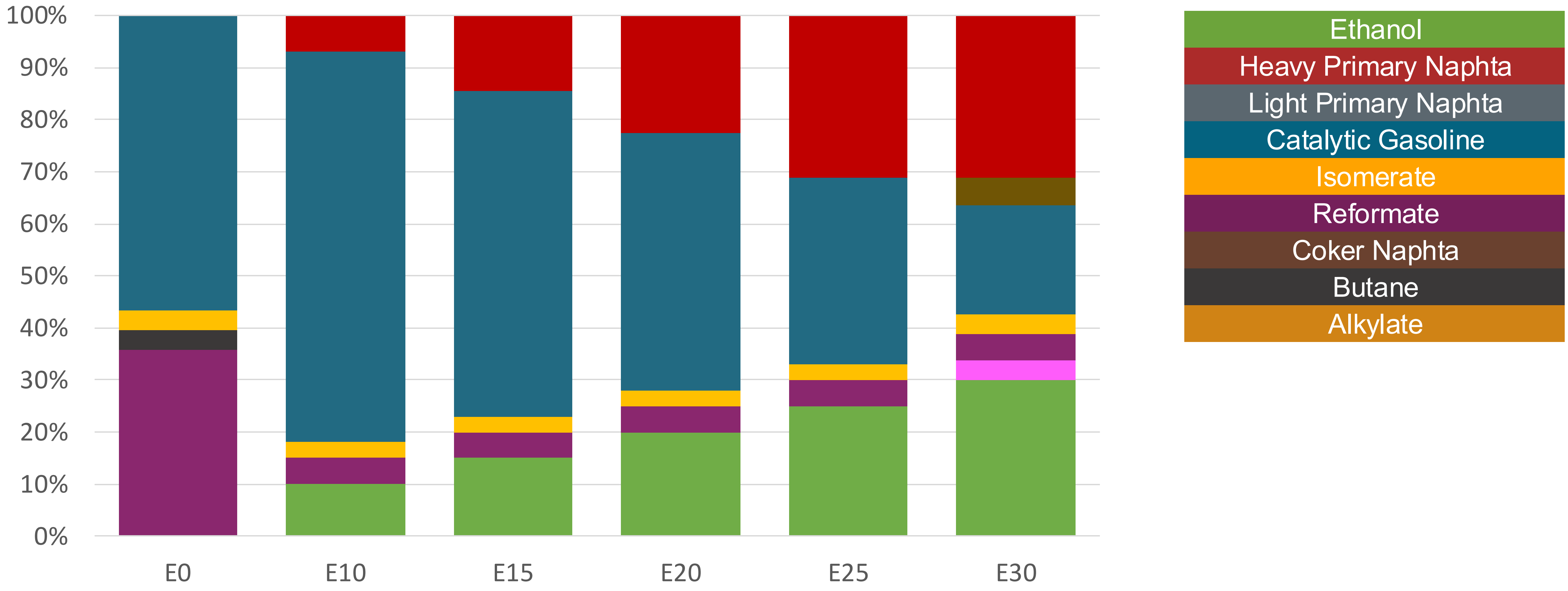
Source: Faro90

Costa Rica – Regular – Constant Octane



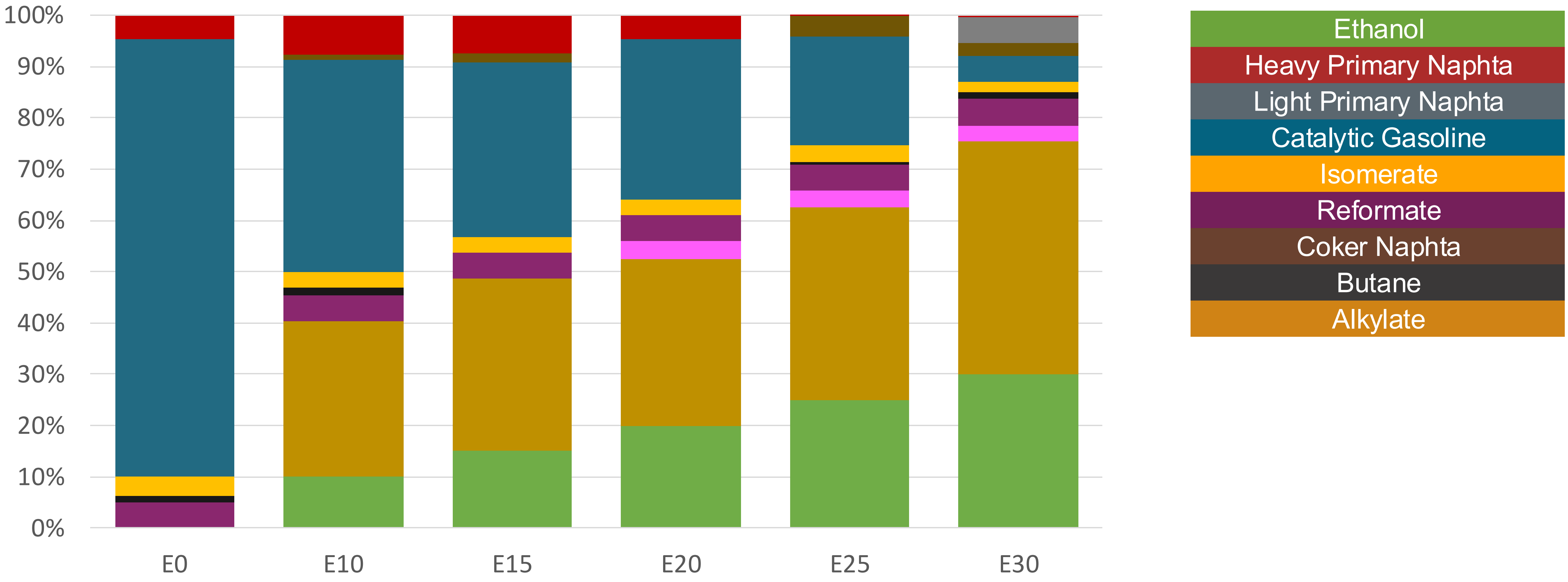
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	$ 2.286	$ 2.028	$ 1.884	$ 1.737	$ 1.586	$ 1.468

Costa Rica – Premium – Constant Octane



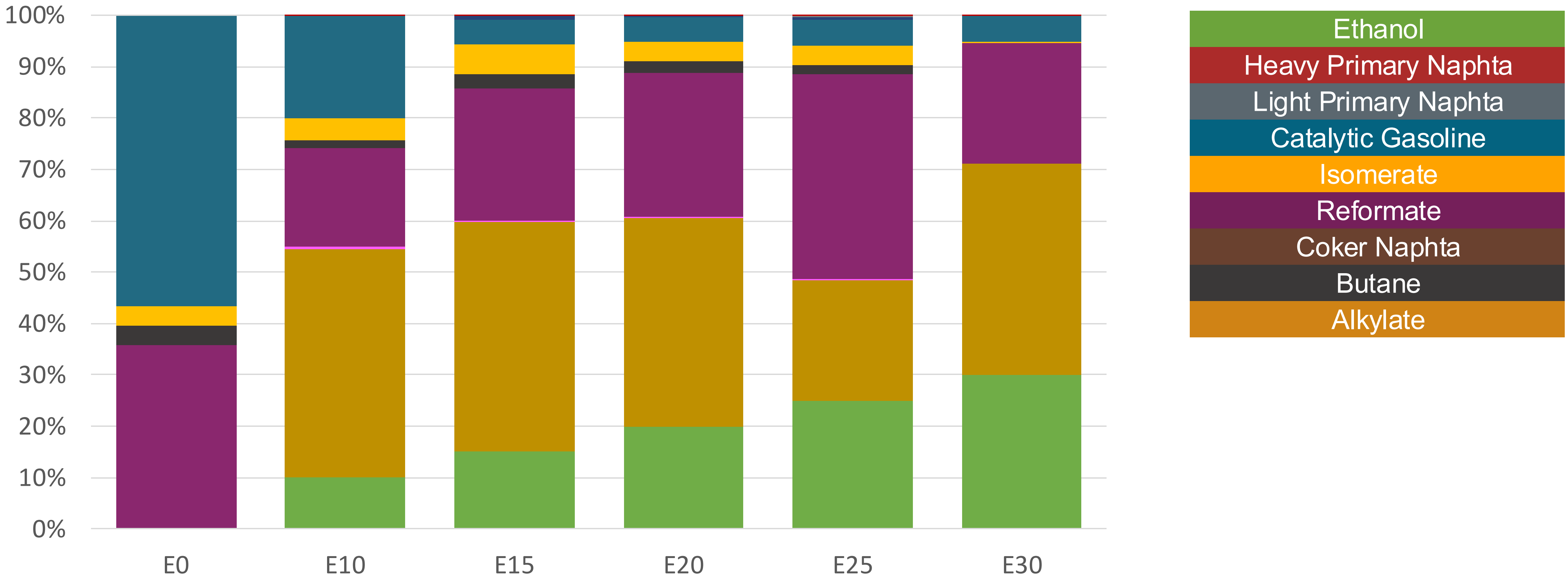
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	$ 2.536	$ 2.233	$ 2.099	$ 1.957	$ 1.807	$ 1.659

Costa Rica – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	$ 2.286	$ 2.286	$ 2.286	$ 2.286	$ 2.286	$ 2.286

Costa Rica – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	$ 2.536	$ 2.536	$ 2.536	$ 2.536	$ 2.536	$ 2.536

Vehicle Emission Impact for Ethanol Gasoline Blending

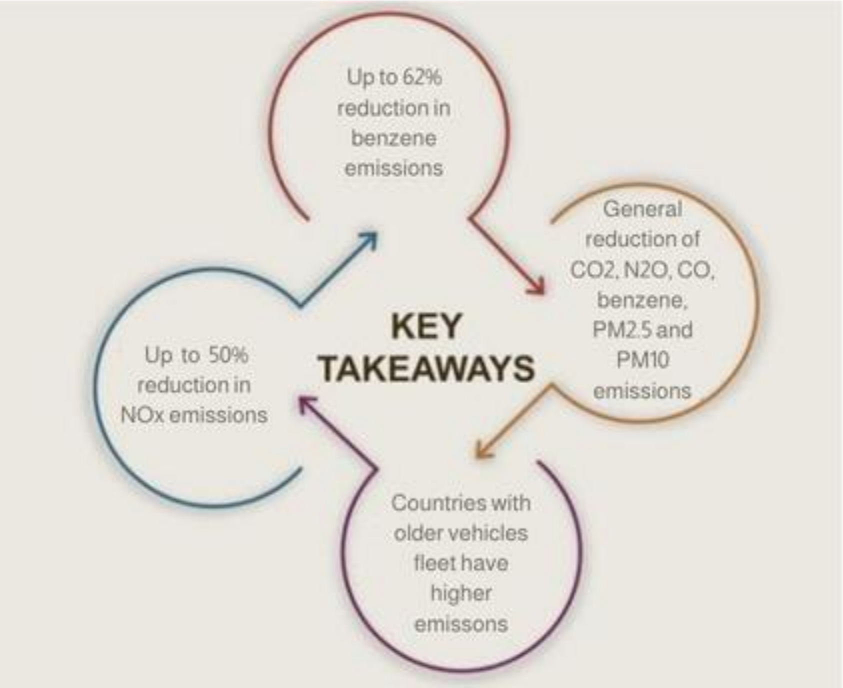
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

Gasoline Vehicular Fleet in Costa Rica

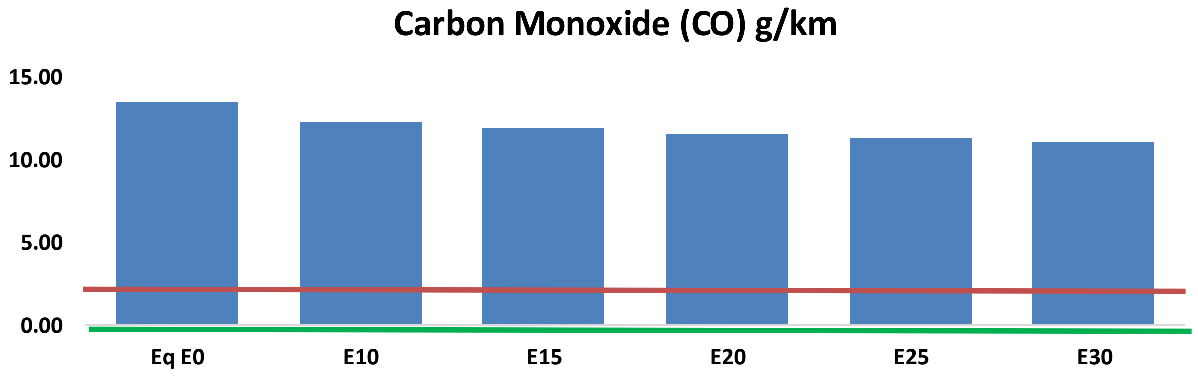
Type	Motor	Exhaust	Recuperator	Age	Number of Vehicles
Auto/Light-Medium	Multi FI-Pt	Euro V	PCV Tank	>4years, >80 mkm	177,395
				4-8 years, 80-160 mkm	201,973
				<8 years, < 160 mkm	135,031
		Euro IV		<8 years, < 160 mkm	132,105
		Euro III		<8 years, < 160 mkm	125,379
		3-Ways		<8 years, < 160 mkm	278,379
Small Engines	FI 4-cycles	Euro III	PCV	>4years, >80 mkm	52,143
				4-8 years, 80-160 mkm	29,684
		Euro II		4-8 years, 80-160 mkm	29,684
				<8 years, < 160 mkm	197,199
Trucks	FI	Euro V	PCV	>4years, >80 mkm	36,587
				4-8 years, 80-160 mkm	41,656
				<8 years, < 160 mkm	27,850
		Euro IV		<8 years, < 160 mkm	27,246
		Euro III		<8 years, < 160 mkm	25,859
		3-Ways		<8 years, < 160 mkm	57,415

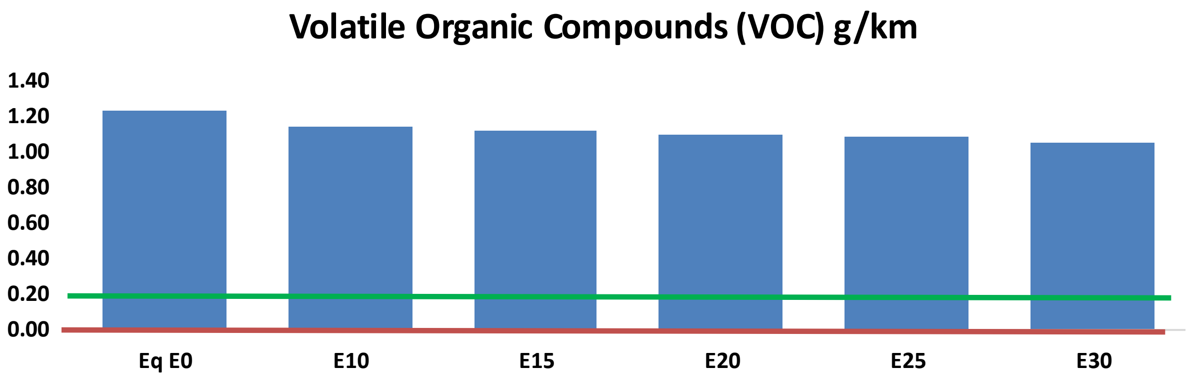
Vehicular Fleet: **1,575,585**
Average Age: **11 years**
Motorcycles: **19.6%**

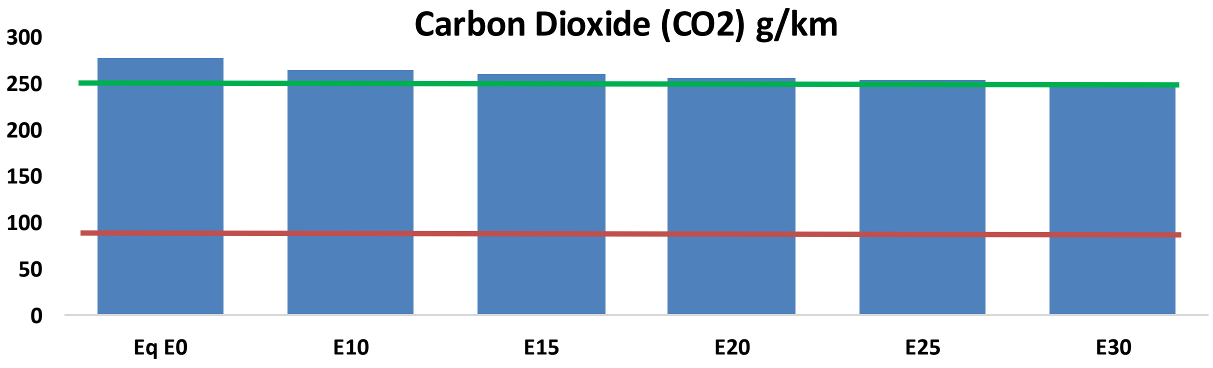
Source: INEC , Faro 90

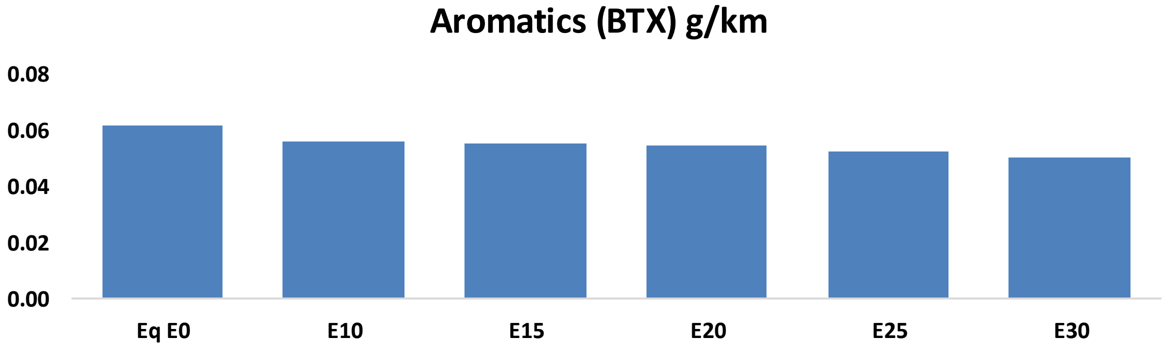
Costa Rica – Vehicular Emissions

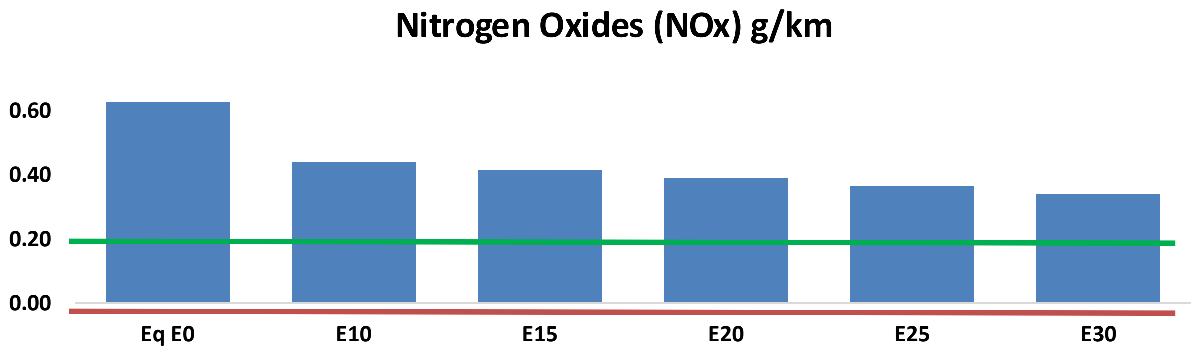
United States
Euro 6

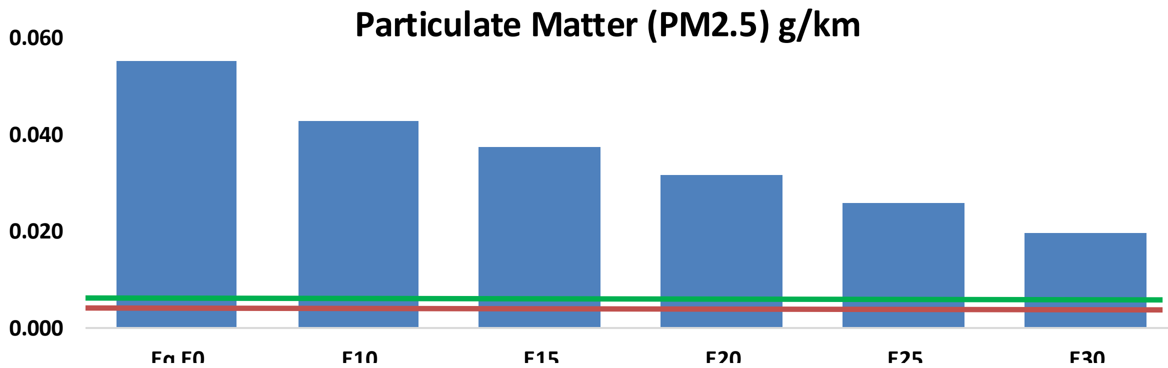












Source: Faro90

Costa Rica – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	497,179	468,254	439,330	417,442	396,250	380,207	359,858	-12%	-20%
CO2	11,924,378	11,617,751	11,328,159	11,100,403	10,987,701	10,923,615	10,722,262	-5%	-8%
VOC	46,610	45,289	43,968	43,328	42,886	42,608	42,020	-6%	-8%
NOx	32,992	24,744	23,094	21,767	20,528	19,162	17,717	-30%	-38%
SOx	143	132	121	112	104	94	84	-15%	-28%
NH3	4,187	4,146	4,104	4,124	4,170	4,203	4,230	-2%	0%
N2O	167	166	165	166	169	171	173	-1%	2%
CH4	9,847	9,847	9,847	9,995	10,211	10,398	10,576	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	378	368	357	352	348	346	341	-6%	-8%
Acetaldehyde	784	900	1,320	2,010	2,738	3,129	3,698	68%	249%
Formaldehyde	2,383	2,317	2,701	3,171	3,323	3,676	4,001	13%	39%
Benzene	838	805	763	751	745	712	689	-9%	-11%
PM2.5	739	656	573	498	423	343	260	-22%	-43%
PM10	833	740	646	561	477	387	293	-22%	-43%

Source: Faro90

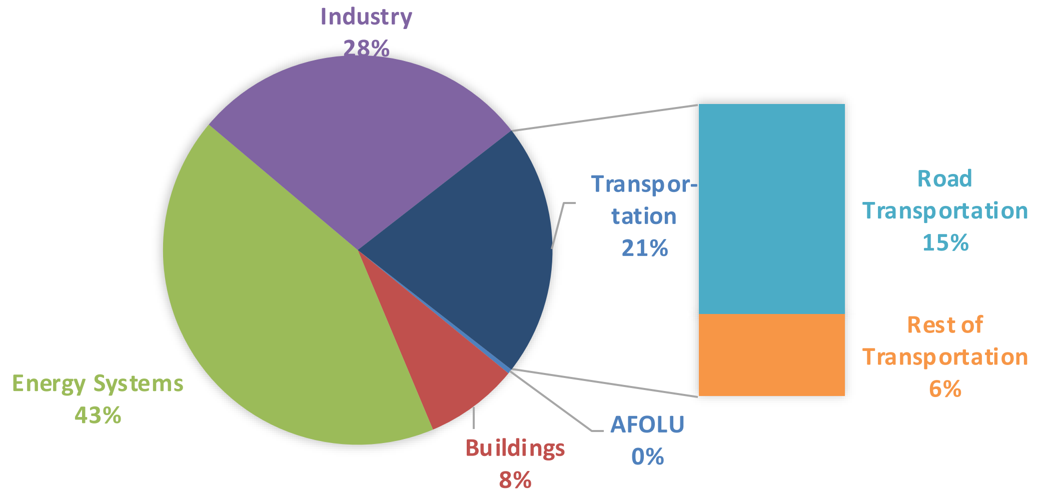
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

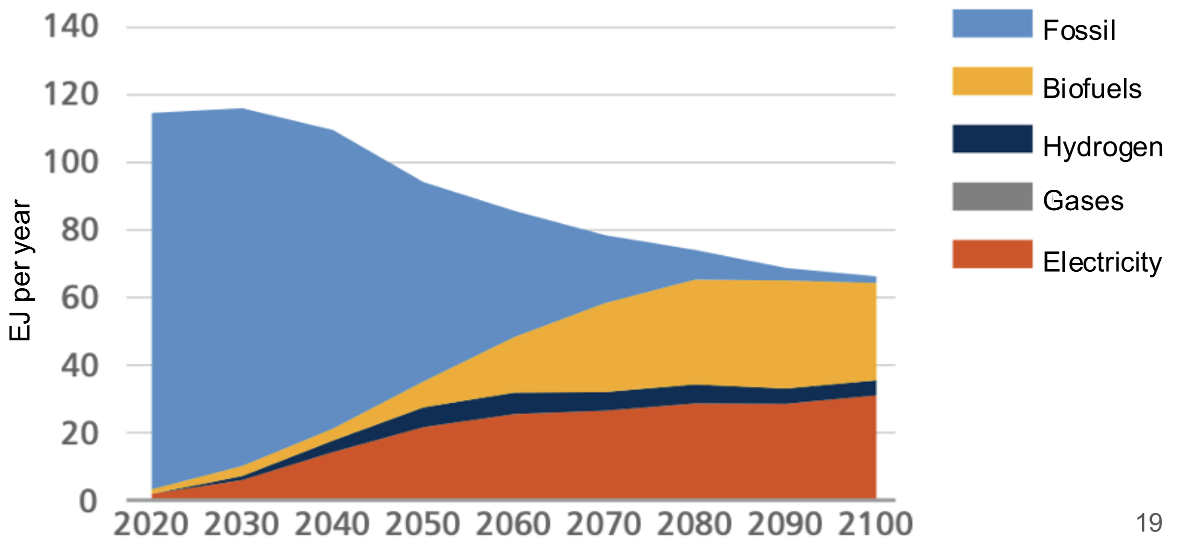
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

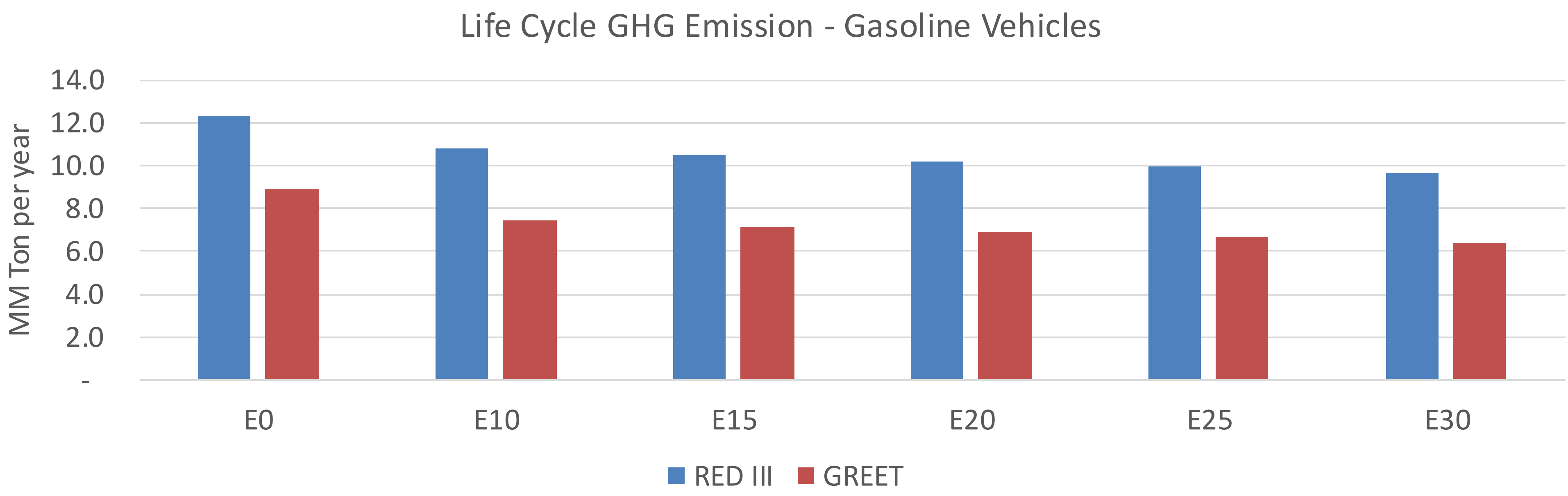
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Costa Rica – Gasoline Vehicle Fleet GEI Emissions



Emisiones País 2024 (MMT/año)	17.1
Meta a 2035 (MMT/año)	8.5
Delta	8.6

%Reducción vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	16.1%	19.4%	22.2%	25.0%	28.5%
RED II/III	0.0%	12.2%	14.8%	17.1%	19.3%	22.0%

% Meta@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	16.7%	20.1%	23.0%	25.9%	29.5%
RED II/III	0.0%	17.5%	21.3%	24.5%	27.7%	31.6%

Source: Greet, RED II/III, climateactiontracker.org, F90